

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset3_expanded • Generated: 2026-04-14 21:02:11

I. CASE IDENTIFICATION	
Dataset ID	dataset3_expanded
Scan ID	ds-003-exp
Source File	windows_soc_data_3_expanded.csv
Primary Threat	Multi-Vector APT Campaign
MITRE Technique	T1059.001
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	120
Aggregate Risk Score	9,700
Threat Level	CRITICAL

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I EXECUTIVE SUMMARY

- A highly sophisticated multi-vector Advanced Persistent Threat (APT) campaign was identified across the environment, impacting 6 hosts and 8 user accounts with a critical risk score of 97%, indicating an active and severe compromise.
- The attack originated primarily through spearphishing attachments, where users executed malicious Office macros, leading to initial system compromise and establishing a foothold within the network.
- Attackers leveraged PowerShell encoded commands extensively to execute payloads, download malicious scripts, and maintain stealthy command execution across multiple systems.
- Persistence mechanisms were established using scheduled tasks and malicious Windows services, ensuring the attackers retained long-term access even after system reboots.
- Privilege escalation was achieved through DLL injection into critical system processes such as svchost.exe, combined with abuse of sensitive privileges like SeDebugPrivilege to gain SYSTEM-level access.
- Defense evasion techniques were actively used, including clearing Windows Event Logs and audit logs, significantly reducing visibility and hindering forensic investigations.
- Credential access was a major focus, with attackers performing LSASS memory dumping using tools like mimikatz and ProcDump, along with SAM registry hive extraction to harvest user credentials and NTLM hashes.
- The attackers successfully utilized Pass-the-Hash techniques and Kerberos ticket requests to perform lateral movement across systems, particularly targeting domain controllers and high-value assets.
- Multiple lateral movement attempts were observed using SMB admin shares, enabling the spread of the attack across different hosts within the network environment.
- Command and Control (C2) communication was established through multiple covert channels, including HTTPS beaconing and DNS tunneling, allowing attackers to remotely control compromised machines.
- Legitimate tools such as certutil were abused to download additional payloads from external C2 servers, demonstrating living-off-the-land techniques to evade detection.
- The attackers deployed additional malicious payloads such as keyloggers through C2 channels, indicating intent to capture sensitive user inputs and credentials over time.
- Data collection activities included large-scale file aggregation using tools like robocopy, followed by archiving sensitive data with password protection using 7zip to prepare for exfiltration.
- Exfiltration of sensitive data was conducted through multiple channels, including FTP transfers and HTTPS POST requests to external attacker-controlled servers, ensuring redundancy in data theft operations.
- Unauthorized access to critical network shares such as SYSVOL indicates potential compromise of domain-level resources and exposure of group policy or authentication data.
- The attack demonstrated a complete kill chain execution, covering all major MITRE ATT&CK; tactics including Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Lateral Movement, Command & Control, Collection, and Exfiltration.
- High resource utilization patterns (CPU, memory, and network I/O) observed during key attack stages further confirm active exploitation, payload execution, and large-scale data movement.
- The coordinated nature of multiple attack chains suggests a well-organized adversary with advanced capabilities, likely aiming for long-term persistence, data exfiltration, and potential future exploitation.
- Immediate incident response actions are required, including isolating affected systems, revoking compromised credentials, blocking C2 communication channels, and performing deep forensic analysis.
- Failure to contain this threat promptly may result in further lateral spread, continued data exfiltration, and severe organizational, financial, and reputational damage.

II CASE IDENTIFICATION & BACKGROUND

Attack Narrative: Lateral Movement via SMB Detection ; Spearphishing to Persistence via Scheduled Task and Service ; DLL Injection and Defense Evasion via Log Clearing ; Credential Harvesting via LSASS and Pass-the-Hash

Hosts	WIN-SOC-PROD-01, WIN-SOC-PROD-02, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-PROD-05, WIN-SOC-DC-01, WIN-SOC-PROD-06
Users	WIN-SOC\Dev-01, NT AUTHORITY\SYSTEM, WIN-SOC\User1, WIN-SOC\Admin, WIN-SOC\Analyst1, WIN-SOC\SvcAcct, WIN-SOC\BackupOp, WIN-SOC\DBAdmin

III DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

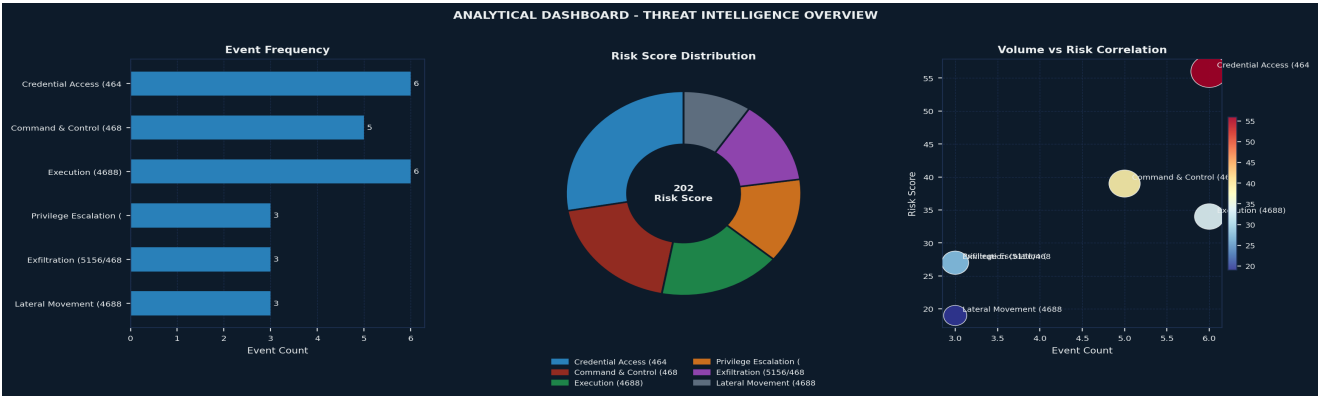


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

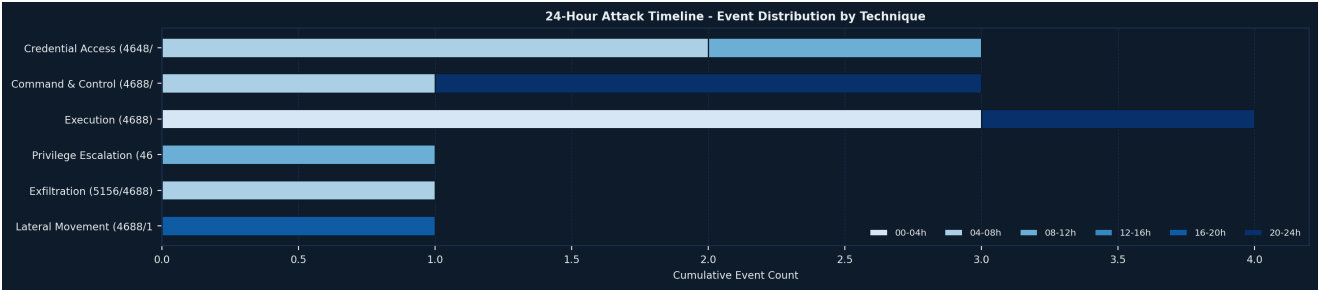


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (4648/10)

LOW

Events: 6	Risk Index: 56	MITRE: T1003.001, T1003.002 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-03-27 03:15:00	WIN-SOC-PROD-05	WIN-SOC\BackupOp	4648
2026-03-27 03:15:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	10
2026-03-27 03:16:00	WIN-SOC-PROD-05	WIN-SOC\BackupOp	4776
2026-03-27 03:17:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688

Analysis: Credential Access (4648/10) produced 6 suspicious artifacts with cumulative risk index 56. The mapped technique T1003.001, T1003.002 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688/1102)

LOW

Events: 2	Risk Index: 18	MITRE: T1070.001, T1078 Tactic: Defense Evasion
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Timestamp	Host	User Context	EID
2026-03-27 02:07:10	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-03-27 02:07:45	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	1102

Analysis: Defense Evasion (4688/1102) produced 2 suspicious artifacts with cumulative risk index 18. The mapped technique T1070.001, T1078 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4688/10)

LOW

Events: 3	Risk Index: 19	MITRE: T1021.002, T1550.002 Tactic: Lateral Movement	
Timestamp	Host	User Context	EID
2026-03-26 23:56:48	WIN-SOC-PROD-01	WIN-SOC\Dev-01	4688
2026-03-26 23:56:49	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	4688
2026-03-27 03:16:45	WIN-SOC-DC-01	WIN-SOC\BackupOp	4768

Analysis: Lateral Movement (4688/10) produced 3 suspicious artifacts with cumulative risk index 19. The mapped technique T1021.002, T1550.002 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Privilege Escalation (4688/8)

LOW

Events: 3	Risk Index: 27	MITRE: T1055.001, T1068 Tactic: Privilege Escalation	
Timestamp	Host	User Context	EID
2026-03-27 02:05:00	WIN-SOC-PROD-04	WIN-SOC\SvcAcct	4688
2026-03-27 02:05:20	WIN-SOC-PROD-04	WIN-SOC\SvcAcct	8
2026-03-27 02:06:00	WIN-SOC-PROD-04	WIN-SOC\SvcAcct	4673

Analysis: Privilege Escalation (4688/8) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1055.001, T1068 under tactic Privilege Escalation indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4698/7045)

LOW

Events: 2	Risk Index: 18	MITRE: T1053.005, T1543.003 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-03-27 01:10:45	WIN-SOC-PROD-03	WIN-SOC\Analyst1	4698
2026-03-27 01:11:30	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	7045

Analysis: Persistence (4698/7045) produced 2 suspicious artifacts with cumulative risk index 18. The mapped technique T1053.005, T1543.003 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-06	9	7	MEDIUM
WIN-SOC-PROD-01	5	1	LOW
WIN-SOC-PROD-03	5	3	LOW
WIN-SOC-PROD-04	5	4	MEDIUM
WIN-SOC-PROD-05	4	3	LOW
WIN-SOC-PROD-02	2	1	LOW
WIN-SOC-DC-01	2	2	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **Lateral Movement via SMB Detection** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

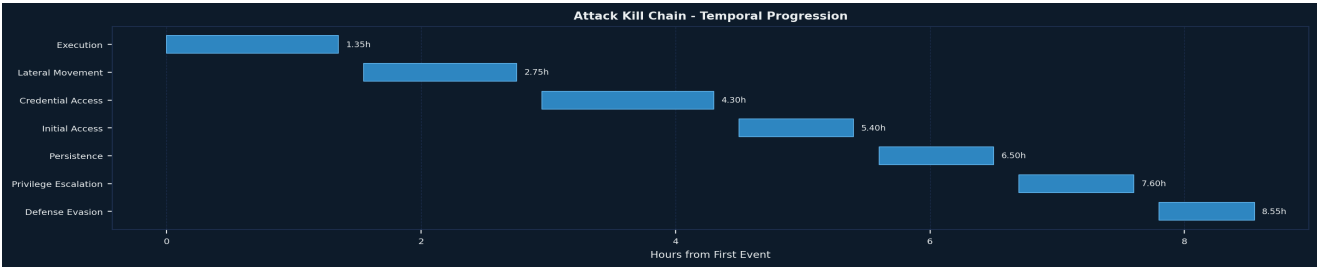
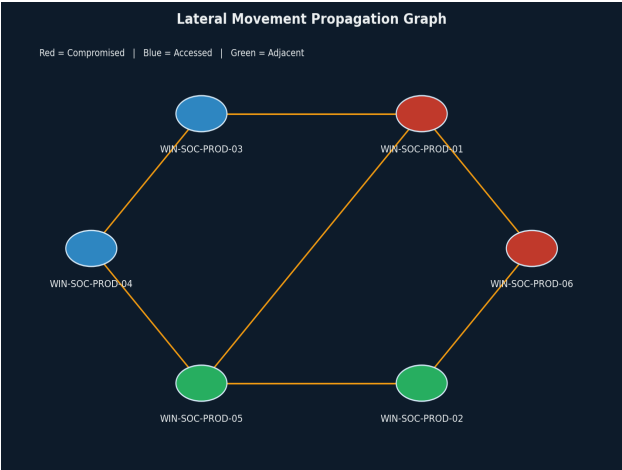


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
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chain-ds-003-01	Encoded IEX Download (PowerShell) → SMB Admin Share Lateral Movement → LSASS Credential Dump (mimikatz.exe)	T1059.001, T1021.002, T1003.001
chain-ds-003-02	Spearphishing Email → Office Macro Execution (winword.exe → cmd.exe) → Scheduled Task Creation → Malicious Service Install	T1566.001, T1059.001, T1053.005
chain-ds-003-03	DLL Injection into svchost.exe → Remote Thread Creation → SeDebugPrivilege Abuse → Windows Event Log Cleared → Audit Logs Deleted	T1055.001, T1068, T1070.001
chain-ds-003-04	Explicit Credential Logon → ProcDump LSASS Memory → NTLM Hash Extracted → Pass-the-Hash Kerberos TGT → SAM Hive Dump	T1003.001, T1003.002, T1550.002
chain-ds-003-05	HTTP C2 Beacon → Outbound Connection to 185.220.101.42 → DNS Tunneling → Certutil Payload Download → Keylogger Dropped	T1071.001, T1071.004, T1105
chain-ds-003-06	Bulk File Copy to Staging → Files Archived with Password (7zip) → FTP Exfiltration to 91.108.4.10 → HTTPS POST Exfil → SYSVOL Share Access	T1074.001, T1560.001, T1041

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	mimikatz.exe	LSASS Dump
Event ID	10	Sysmon
Process	winlogon.exe	Suspicious Phishing Email Opened - Macro Executed
Event ID	4625	Security
Process	winword.exe	Office Macro Spawned cmd.exe
Process	schtasks.exe	Malicious Scheduled Task Created for Persistence
Event ID	4698	Security
Process	services.exe	Malicious Service Installed for Persistence
Event ID	7045	System
Process	lsass.exe	Logon Success After Phishing
Event ID	4624	Security

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

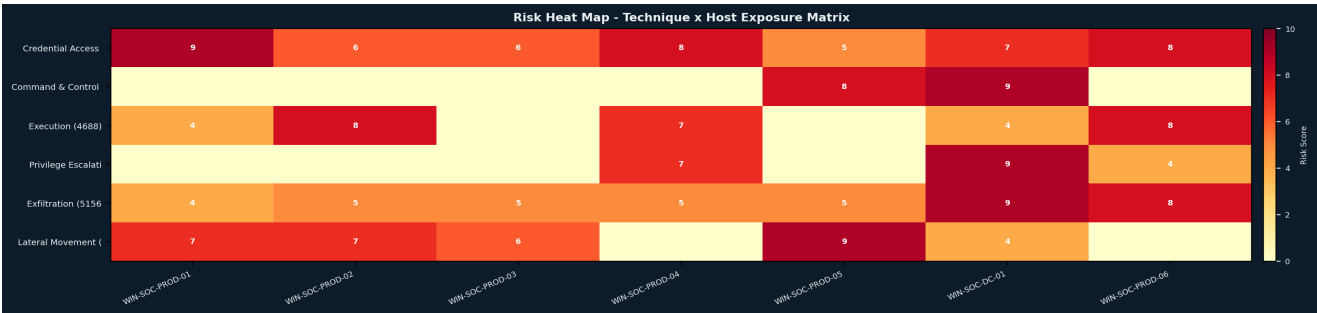


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

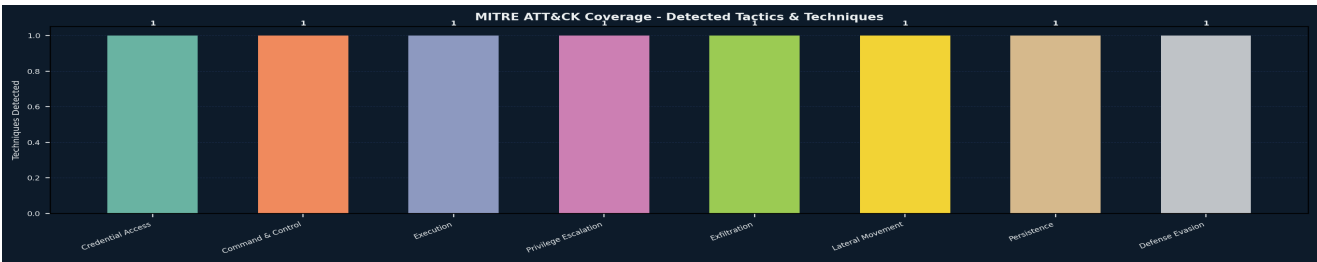


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.001, T1003.002	Credential Access (4648/10)	Credential Access
T1071.001, T1071.004	Command & Control (4688/5156)	Command & Control
T1059.001	Execution (4688)	Execution
T1055.001, T1068	Privilege Escalation (4688/8)	Privilege Escalation
T1041	Exfiltration (5156/4688)	Exfiltration
T1021.002, T1550.002	Lateral Movement (4688/10)	Lateral Movement
T1053.005, T1543.003	Persistence (4698/7045)	Persistence
T1070.001, T1078	Defense Evasion (4688/1102)	Defense Evasion

XVII

REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

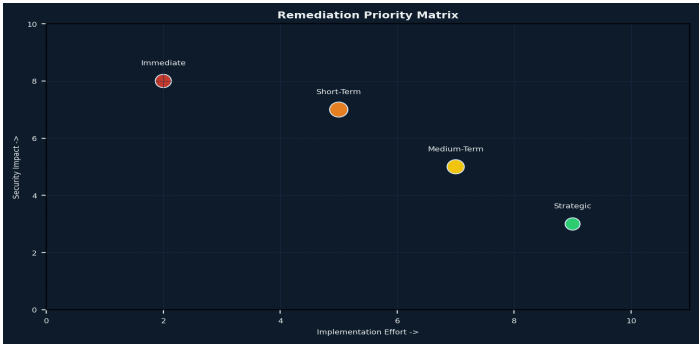


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII

EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset3_expanded**, the environment exhibits a **CRITICAL** severity posture with aggregate risk index **9,700** across **10,000** records. Dominant indicators include **Credential Access (4648/10) (6 events)**, **Command & Control (4688/5156) (5 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst

Automated Forensic Engine

Authorizing Officer

Incident Response Division

XIX

ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 21:02:17

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ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-03-26 23:56:48	WIN-SOC-PROD-01	WIN-SOC\Dev-01	powershell.exe	Encoded IEX Download
2026-03-26 23:56:49	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-03-26 23:56:50	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-26 23:56:51	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-26 23:56:52	WIN-SOC-PROD-01	WIN-SOC\Admin	powershell.exe	Encoded IEX Download
2026-03-26 23:57:38	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-26 23:57:52	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-27 01:10:05	WIN-SOC-PROD-03	WIN-SOC\Analyst1	winlogon.exe	Suspicious Phishing Email Opened - Macro Executed
2026-03-27 01:10:20	WIN-SOC-PROD-03	WIN-SOC\Analyst1	winword.exe	Office Macro Spawned cmd.exe
2026-03-27 01:10:45	WIN-SOC-PROD-03	WIN-SOC\Analyst1	schtasks.exe	Malicious Scheduled Task Created for Persistence
2026-03-27 01:11:30	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	services.exe	Malicious Service Installed for Persistence
2026-03-27 01:12:10	WIN-SOC-PROD-03	WIN-SOC\Analyst1	lsass.exe	Logon Success After Phishing
2026-03-27 02:05:00	WIN-SOC-PROD-04	WIN-SOC\SvcAcct	rundll32.exe	DLL Injection into svchost.exe
2026-03-27 02:05:20	WIN-SOC-PROD-04	WIN-SOC\SvcAcct	svchost.exe	Remote Thread Created in svchost.exe
2026-03-27 02:06:00	WIN-SOC-PROD-04	WIN-SOC\SvcAcct	svchost.exe	SeDebugPrivilege Abuse
2026-03-27 02:07:10	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	wevtutil.exe	Windows Event Logs Cleared
2026-03-27 02:07:45	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	wevtutil.exe	Audit Log Cleared
2026-03-27 03:15:00	WIN-SOC-PROD-05	WIN-SOC\BackupOp	lsass.exe	Explicit Credential Logon Attempt
2026-03-27 03:15:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	procdump.exe	LSASS Memory Dump via ProcDump
2026-03-27 03:16:00	WIN-SOC-PROD-05	WIN-SOC\BackupOp	lsass.exe	NTLM Hash Extracted and Used

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
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EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.